



C# Project Documentation

Library Management System

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1. Introduction

Purpose

The **Library Management System** is designed to streamline and automate the core operations of a library, replacing error-prone manual processes with a digital solution. Its primary goal is to simplify tasks such as book cataloging, student management, and transaction tracking, enabling librarians and staff to manage resources efficiently, reduce administrative workload, and enhance service delivery.

Scope

The system provides the following key functionalities:

- **Book Management:**
 - Add, update, delete, and search books by details (title, author, ISBN).
 - View real-time availability and status (issued/available).
 - Issue books to students and track return deadlines.
- **Student Management:**
 - Add or remove student profiles.
 - View student details.
- **User-Friendly Interface:** A Windows Forms-based GUI for intuitive navigation.
- **Secure Access:** Role-based login for librarians and staff.

By centralizing these operations, the system minimizes manual record-keeping, accelerates book search and retrieval, and ensures accurate tracking of transactions.

Target Audience

This application is tailored for:

- **Librarians and Library Staff:** To manage books, oversee student registrations, and monitor borrowing activities and to assist in daily operations such as issuing/returning books /.

The system empowers users to focus on improving library services rather than administrative tasks, fostering a seamless experience for the staff.

2. System Overview

Technology Stack

- **Frontend:**
 - Language: C#
 - Framework: .NET Windows Forms
 - IDE: Visual Studio Community 2022
- **Backend:**
 - Database: Microsoft SQL Server
 - Tools: SQL Server Management Studio (SSMS) for database design and queries.
- **Connectivity:** ADO.NET for seamless interaction between the C# application and SQL database.

Architecture

The system follows a **client-server architecture**:

1. **Presentation Layer (Client):**
 - Built with Windows Forms for a user-friendly GUI.
 - Handles user input (e.g., adding books, issuing to students).
2. **Database Layer (Server):**
 - SQL Server stores all data (books, students, transactions).

Key Components

- **Modules:**
 - **Book Management:** Add, update, delete, search, view books, and Issue/return books with due dates.
 - **Student Management:** Add, remove, and view student records.

Deployment Environment

- **Development:**
 - Built and tested on Windows OS.
 - Database hosted locally on SQL Server.

3. Detailed Features and Screenshots

3.1. Login Page

Description

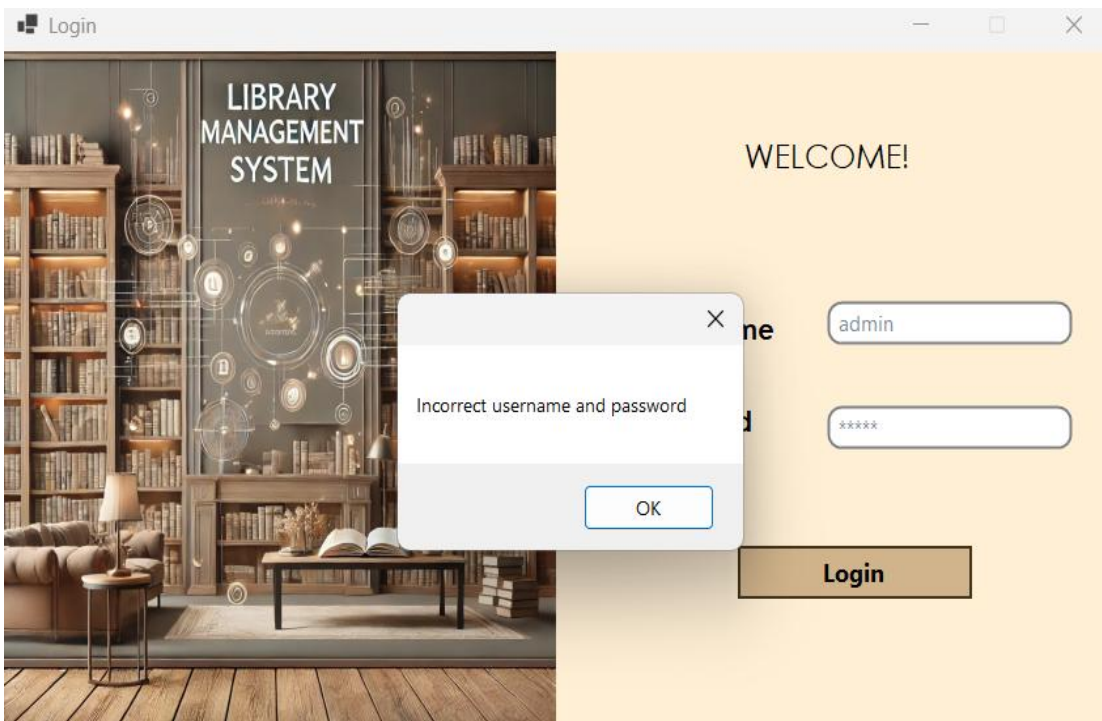
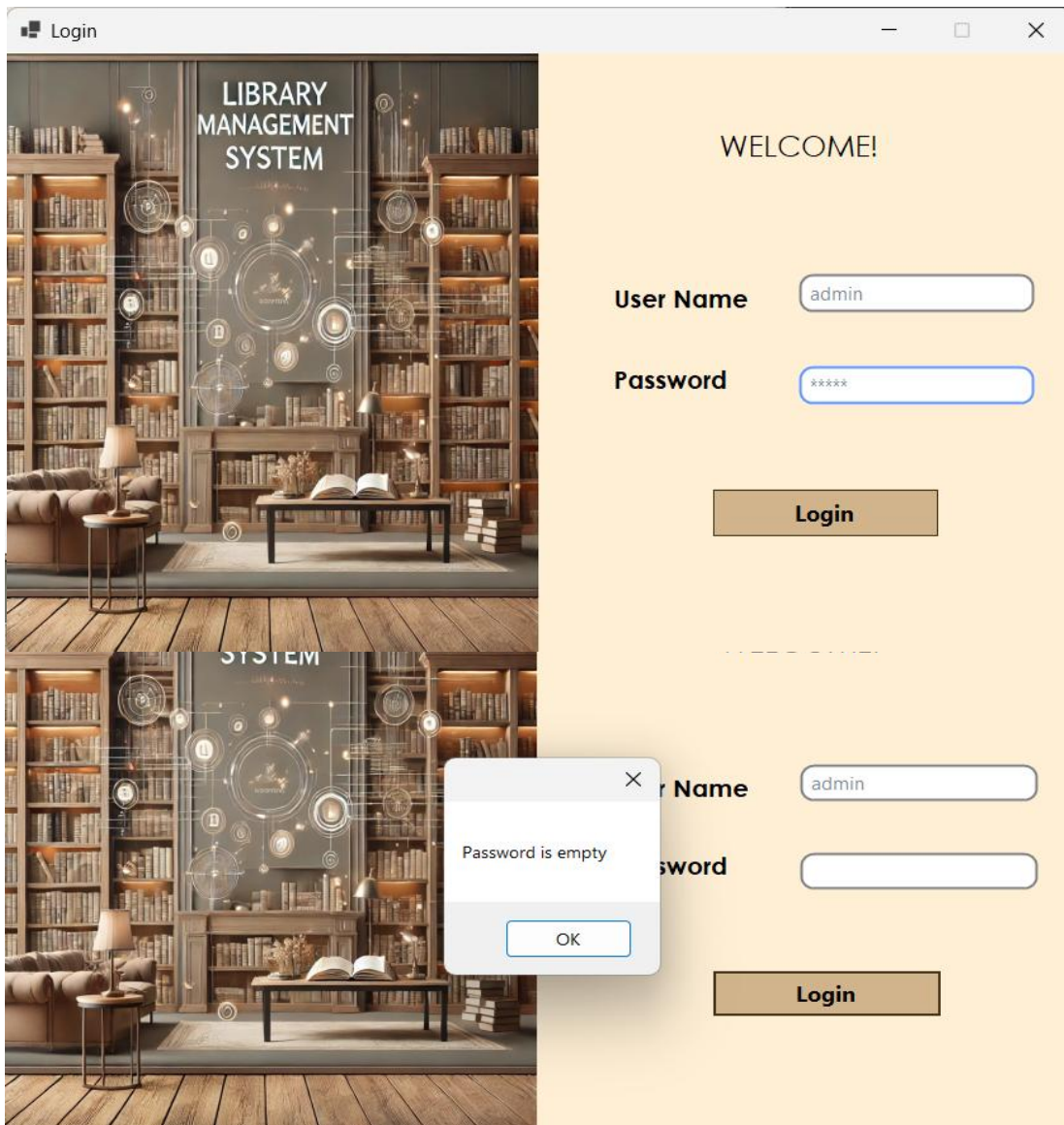
The login page ensures secure access to the system by authenticating authorized users (librarians and staff). Key functionalities include:

1. **Input Validation:** Checks for empty fields and displays errors.
2. **Authentication:** Validates credentials against the database.
3. **Caps Lock Helper:** Shows a warning if Caps Lock is active during password entry.

UI Components Used

1. **Textboxes:** For username and password input.
2. **Buttons:**
 - *Login:* Triggers authentication.
 - *Exit:* Closes the application.
3. **Panel:** Groups login elements for a clean layout.
4. **Picture Box:** Displays a logo or library-themed image.
5. **Error Labels:** Shows validation messages (e.g., "Please enter both fields!").

Screenshots



3.2 Dashboard

Description

The dashboard serves as the central hub for librarians and staff to access all system functionalities. It features:

1. **Animated Sidebar Menu:**

- A collapsible sidebar with dropdown options for **Manage Books** and **Manage Students**.
- Smooth animations (using Timer controls) to expand/collapse menus when buttons like the "3-dash" (≡) or management buttons are clicked.

2. **Role-Based Navigation:**

- **Manage Books:** Dropdown includes *Add Books*, *View Books*, *Issue Book*, *Book Details*, *Update Book*, *Return Book*.
- **Manage Students:** Dropdown includes *Add Student*, *Update Student*, *View Student*.

3. **User Actions:**

- **My Profile:** View/edit logged-in user details.
- **Logout:** Securely logs out and redirects to the login page.

UI Components Used

1. **Panels:**

- `panelMenu`: Hosts the sidebar and dropdown options.
- `panelSubMenuBooks`/`panelSubMenuStudents`: Contain sub-options (e.g., "Add Books").

2. **FlowLayoutPanel**: Automatically arranges sub-buttons vertically.

3. **Buttons:**

- **≡ (3-dash)**: Toggles sidebar expansion.
- **Manage Books/Students**: Triggers dropdown animations.
- **Sub-options** (e.g., "View Books"): Navigate to specific forms.

4. **PictureBox**: Displays icons (e.g., book/student icons).

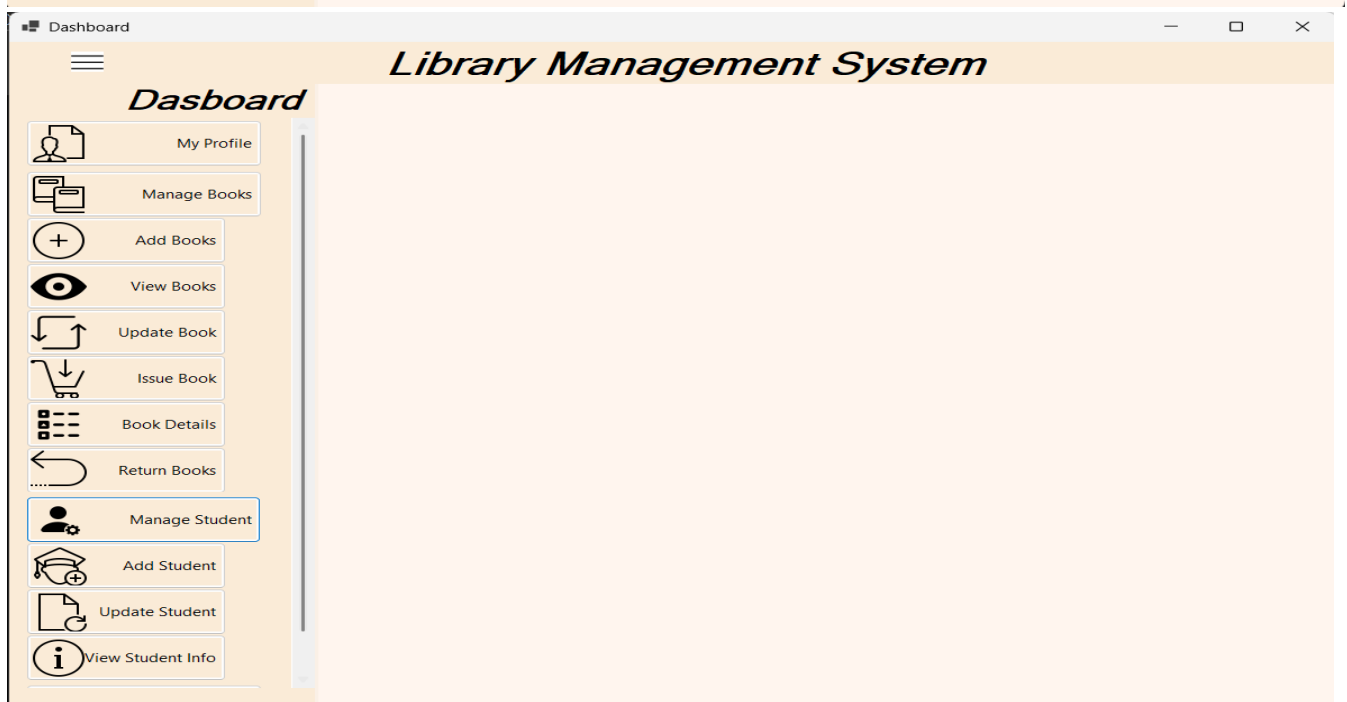
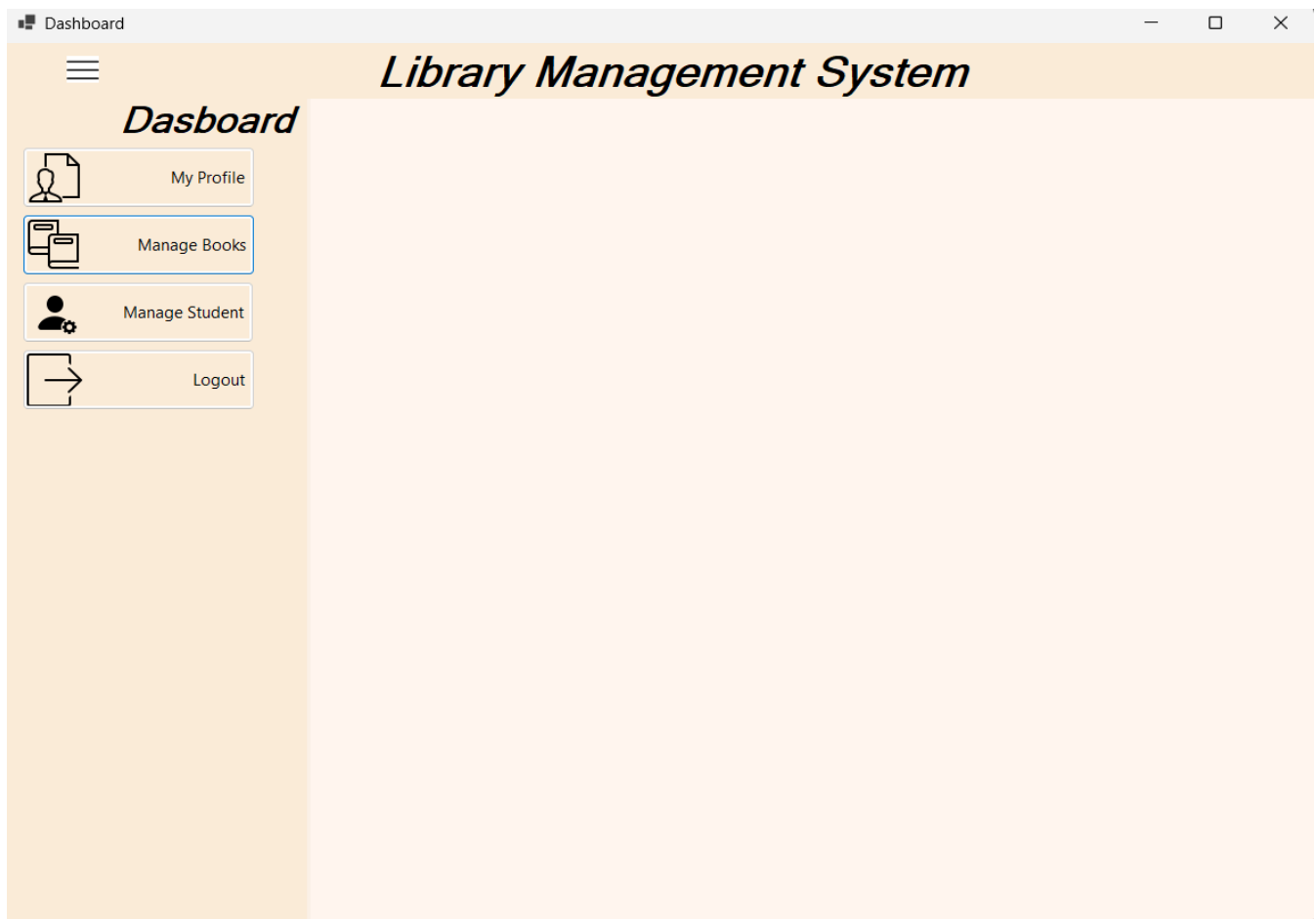
5. **Labels**: section titles.

6. **Timers:** Control smooth animations for menu transitions.

Key Features

- **Responsive Design:** Menus expand/collapse smoothly for a modern user experience.
- **Role-Based Access Control:** Tailored options for admin/staff.
- **Intuitive Navigation:** Sub-buttons are grouped logically under parent categories.

Screenshots:



3.2.1 My Profile

Description

The **My Profile** section displays detailed information about the logged-in user and provides role-specific actions. Key features include:

1. **User Details:** Shows static fields (ID, name, email, salary, date of birth, username, admin status).
2. **Role-Based Options:**
 - **Admin Users:** Can *Add Staff*, *Update Staff*, *View Staff Details*, and *Change Password*.
 - **Non-Admin Users:** Limited to *Change Password*.
3. **Close Button:** Exits the profile section and returns to the dashboard.

UI Components Used

1. **Labels:** Display user details (ID, name, email, etc.).
2. **Buttons:**
 - **Admin Actions:** *Add Staff*, *Update Staff*, *View Staff*.
 - **Change Password:** Opens password update form.
 - **Close:** Exits the profile section.
3. **DateTimePicker:** input user's date of birth.
4. **Panels:** Organize sections (e.g., user info, action buttons).
5. **Radio Buttons:** Optional (e.g., to toggle admin status during updates).

Key Features

- **Dynamic Interface:** UI adapts based on the logged-in user's role.
- **Security:** Password changes require confirmation of the old password.
- **Readability:** Clean layout with clearly labeled fields and buttons.

Screenshots

Add_staff

Add Staff

First Name

Last Name

Email

Salary

Date of birth

Tuesday , February 4, ▾

Admin

☐ Yes

Save

Clear

Cancel

update_staff

—

□

×

Update Staff

ID Number

First Name

Last Name

Email

Salary

Date_of_birth

Admin

☐ Yes

Update

Delete

Clear

Cancel

view_staff

View Staff

Staff ID

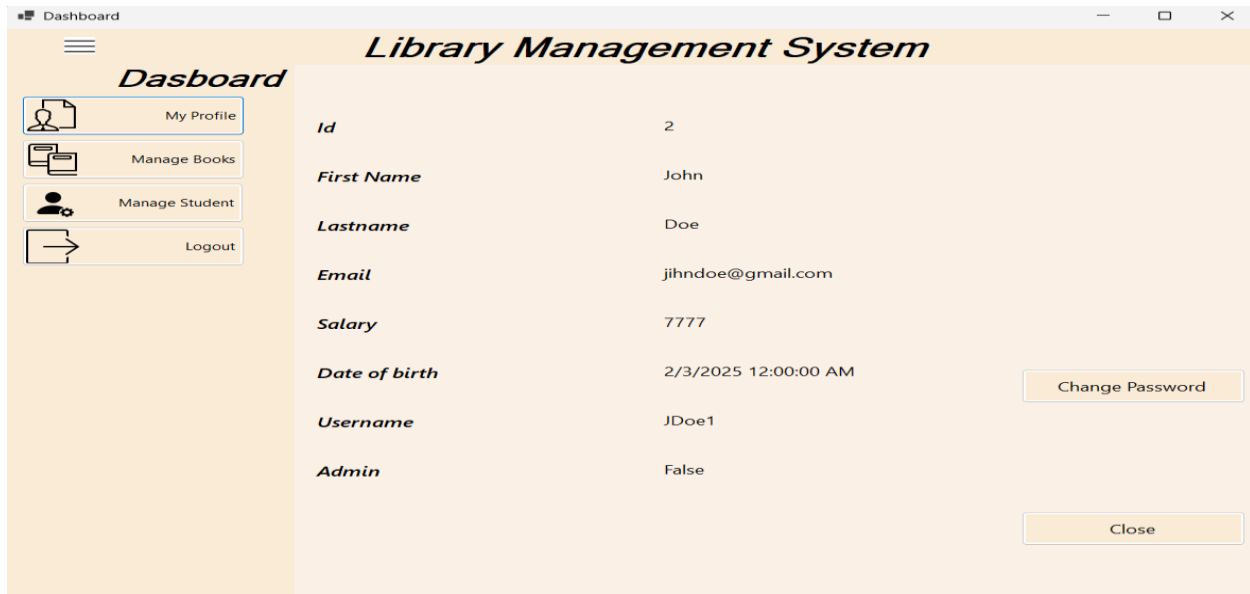
change_password

Previous password

New password

Confirm Password

For non admins:



3.2.2 Manage Books

The **Manage Books** module allows librarians to perform all book-related operations. Users can access the following functions via dropdown buttons:

1. **Add Books:** Register new books into the system by entering details (title, author, ISBN, etc.).
2. **View Books:** Display a list of all books with filters or search by **Book ID**.
3. **Issue Book:** Assign a book to a student by searching for the **Book ID** and linking it to a student ID.
4. **Book Details:** View comprehensive information about a specific book (e.g., status, issue history).
5. **Update Book:** Modify book details (e.g., title, author) and delete books using the **Book ID** as a reference.
6. **Return Book:** Mark a book as "available" by searching for the **Book ID** and updating its status.

All workflows are driven by Book ID to ensure accuracy and avoid duplicates.

UI Components Used

1. **Buttons:**
 - Sub-options (Add, View, Issue, Update, Return).
 - Search, Save, Cancel.
2. **Textboxes:** For entering Book ID, title, author, etc.
3. **DataGridView:** Displays search results (e.g., all books).

4. **Labels:** Guide users through workflows (e.g., "Enter Book ID").
5. **Panels:** Organize forms and input sections.
6. **DateTimePicker:** Sets due dates for book issuance.

Key Features

- **Real-Time Status Updates:** Book availability changes instantly after issuance/return.
- **Input Validation:**
 - Prevents empty fields during book addition.
- **Transaction Integrity:** Uses SQL transactions to ensure book status and student records stay synchronized.

Screenshots:

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

Add Books

Book Name

Book Author Name

Book Publication

Tuesday , February 4 ,

Book Purchase Date

Tuesday , February 4 ,

Book Price

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

View Books

Book Name

search

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Logout

Update Book

Book ID

Book Name

Book Author Name

Book Publication

Tuesday , February 4 ,

Book Purchase Date

Tuesday , February 4 ,

Book Price

Search

Update

Delete

Clear

Cancel

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Logout

Issue Books

Student Id

Student Name

Department

Student Phone Number

Student Email

Book ID

Book Due Date

Tuesday , February 4 ,

Search

Issue Book

Clear

Cancel

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Logout

Issued Books

Refresh

Returned Books

Refresh

Dashboard

Library Management System

Dashboard

My Profile

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Logout

Return Book

Student Id

Student Name

Department

Student Phone Number

Student Email

Book ID

Search

Return Book

Clear

Cancel

3.2.3 Manage Students

Description

The **Manage Students** module enables librarians to handle student registrations and updates. Key functionalities include:

1. **Add Students:** Register new students by entering details (name, ID, email, enrollment date, etc.).
2. **View Students:** Display a list of all students or search for a specific student by **Student ID**.
3. **Update Students:** Modify student information (e.g., name details) after searching by **Student ID** also has a delete functionality.

All operations rely on Student ID to ensure accurate record management.

UI Components Used

1. **Buttons:**
 - Sub-options: *Add Student, View Students, Update Student*.
 - *Search:* Triggers Student ID lookup.
 - *Save/Cancel:* For adding/updating records.
2. **Textboxes:** Input Student ID, name, email, etc.
3. **DataGridView:** Displays student records in tabular format (View Students).
4. **Labels:** Guide users (e.g., "Enter Student ID").
5. **Panels:** Organize input sections and forms.
6. **DateTimePicker:** Select enrollment/update dates.

Key Features

- **Real-Time Updates:** Changes reflect immediately in the database and UI.
- **Input Validation:**
 - Prevents duplicate Student IDs during registration.
 - Checks for empty fields or invalid formats (e.g., email).
- **Student ID-Centric Workflow:** Ensures precision in all operations.

Screenshots:

Dashboard

Library Management System

Dashboard

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Add Student

Add Student

Student Name

ID Number

Department

Student Phone Number

Student Email

Save

Clear

Cancel

Dashboard

Library Management System

Dashboard

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Add Student

Update Student

View Student Info

Logout

Update Student

ID Number

Search

Student Name

Department

Student Phone Number

Student Email

Update

Clear

Cancel

Delete

Dashboard

Library Management System

Dashboard

Manage Books

Add Books

View Books

Update Book

Issue Book

Book Details

Return Books

Manage Student

Add Student

Update Student

View Student Info

Logout

View Student

ID Number

search

3.2.4 Logout

Description

The Logout feature securely terminates the current user session and redirects the user back to the login page. It ensures no unauthorized access to the system after logging out.

4.Summary

The **Library Management System** is a robust Windows Forms application developed using **C#** and **SQL Server** to automate and streamline library operations. Designed for efficiency and user-friendliness, the system empowers librarians and staff to manage books, students, and transactions seamlessly.

Key Highlights:

1. Core Features:

- **Book Management:** Add, view, update, issue, return, and search books by **Book ID**.
- **Student Management:** Register, update, and track students using **Student ID**.
- **Authentication:** Secure login with Caps Lock warnings, input validation, and role-based access (admin/staff).
- **Dashboard:** Animated sidebar with dropdown menus for intuitive navigation.

2. Technology:

- Built with **C# Windows Forms** (.NET Framework) for the frontend.
- **SQL Server** for backend database operations.
- **ADO.NET** for database connectivity and real-time updates.

3. User-Centric Design:

- Smooth animations (using Timer controls) for menu transitions.
- Input validation, error handling, and role-specific interfaces (e.g., admin privileges).
- Clear workflows for issuing/returning books and updating records.

4. Impact:

- **Efficiency:** Reduces manual work by automating book tracking and student management.
- **Accuracy:** Ensures data integrity through **Book ID** and **Student ID**-based operations.
- **Security:** Password protection, role-based access, and secure database interactions.

This system modernizes traditional library workflows, offering a scalable and reliable solution for educational institutions or public libraries.